When Better Choice Models Do Not Matter

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Abstract

Before making a model more expressive, it is worth asking whether the errors it is meant to fix are the errors the system actually makes. We develop that question into a working method for preference ranking. The idea is to compute, before the comparison is fit, an *effect ceiling*: the largest improvement any richer model in a candidate family could deliver, given where the data actually live. The ceiling then anchors an *error budget* that assigns the system's observed error to approximation, estimation, and drift. We apply the method to a question the community keeps raising about Chatbot Arena, which ranks language models with Bradley–Terry despite well known objections to its independence assumption. A richer Thurstonian alternative with flexible noise and native tie handling seems like it should help. The ceiling says otherwise: on the ability gaps real matchups occupy, the two model families never disagree by more than about one percentage point of win probability, so almost no improvement was available to find. A pre-registered audit of 1.67 million votes, replicated on a second release, confirms the ceiling on the decisive channel, and the one channel that exceeds its ceiling, ties, does so through drift rather than link shape. What actually limits the system is estimation error, cold-start coverage, and nonstationary drift; the two that admit the ceiling's unit exceed it by one to two orders of magnitude, and cold-start coverage is not expressible in that unit at all. Our evidence covers one platform and one family of links, so we state the broader lesson as a conjecture the case study supports rather than proves: expressiveness pays only where approximation error is the binding constraint, and the ceiling tells you in advance whether it is.

1 Introduction

A recurring situation in applied modeling goes like this. A deployed system uses a simple model. The simple model rests on an assumption that theory says is wrong. A richer model exists that fixes the assumption. Someone proposes switching. The proposal sounds like progress, and evaluating it sounds straightforward: fit both models and see which one wins.

This paper argues that the comparison should start somewhere else. Before fitting either candidate, one can often compute how large the richer model's advantage could possibly be. The two models are functions, the data occupy a known region, and if the functions barely differ on that region, then no amount of fitting will separate them. We call this computation an effect ceiling. When the ceiling comes out far below the smallest difference that would matter in practice, the comparison is settled before it begins, and attention should move to the errors that remain. Assigning the system's observed error to its actual sources, which we call an error budget, is the second half of the method. Together the two computations answer a question that matters well beyond any single comparison: is this system limited by what its model can express, or by what its data can estimate?

Two boundaries on that question belong up front rather than in the limitations. The method applies when the competing models make comparable predictions from a shared input whose distribution can be observed before the comparison is fit, and when the application admits a defensible answer to how small a difference is too small to matter. Rating systems, reward models, and discrete choice models generally qualify; comparisons that change the prediction target or the feature set do not, and domains without a natural threshold make the framework harder to operationalize, a point Section 9 returns to. The ingredients are also old, and we prefer to say so here. The threshold logic is the logic of equivalence testing from bioequivalence trials [19]. The ceiling is a divergence bound of the kind statistics has long used to say when two hypotheses cannot be told apart [5, 13]. The identity behind it is standard theory of proper scoring rules [10]. What we claim is the assembly: computing these quantities before the comparison is fit, as a design step, and letting them set the expectations an audit then confirms or refutes.

We develop the method on a live case. Chatbot Arena ranks language models by fitting Bradley–Terry to millions of pairwise human votes [2, 3]. Bradley–Terry assumes independence of irrelevant alternatives, an axiom human preferences owe nothing to [15], and the platform's comparison pool grows every month while ties, a fifth of quality votes, sit outside the model's outcome space entirely. The classical remedy is the Thurstonian family [21], made exact for arbitrary noise by Cotton's lattice construction [4], with ties arising naturally as dead heats rather than by parametric patch [6]. The hypothesis that a richer, tie-aware link should rank better writes itself, and to our knowledge nobody had tested it at scale.

The ceiling says the hypothesis never had room to be true. Real matchups concentrate near toss-ups, where every sensible link agrees. On the observed distribution of ability gaps, the best possible advantage of any link in this Thurstonian family over the logistic is at most 0.23 times our minimum practical difference for decisive votes, a threshold we derive from a ten-Elo-point reading error, and the analogous ceiling for the tie mechanisms is at most 0.31 times the separate threshold we derive for tie prediction. At no observed matchup do the two links disagree by more than about one percentage point of win probability. We prove the general version of this argument as Theorem 2: when two model families are uniformly close on the support of the data, no fitting procedure, and in fact no state of the world, can separate them by more than a computable amount.

The audit confirms the ceiling on the decisive channel in every direction we test, and the one measurement that exceeds a ceiling, on the tie channel, does so through drift and a single loud month rather than through link shape, for reasons Section 4 spells out. Across sixteen months of pool growth, refit stability is statistically indistinguishable between the families. Held-out calibration is practically equivalent, with a real but sub-practical lean toward Bradley–Terry. The two tie mechanisms have provably different shapes, yet fit observed ties equally well, because the data are thin exactly where the shapes diverge. Every comparison replicates on an independent 2025 release. Meanwhile the budget shows where the error actually lives: all methods share the same forty-six percent underconfidence on next-month votes, up to three quarters of next-month votes involve a brand-new model that no ability estimator can score, and the tie rate itself drifts upward by nearly three quarters over the study period. The first and the last of these exceed the entire link-choice ceiling by one to two orders of magnitude in its own unit, and the cold-start share is not expressible in that unit at all.

Three findings give the audit its texture. First, the families do differ, but structurally rather than statistically: Davidson's tie parameter provably never touches its decisive predictions, while the lattice's single width parameter moves tie mass and decisive steepness together (Theorem 1); we attempt to price that entanglement, and our first estimate did not survive its own stress test (Section 6.4). Second, the one apparent exception we observed, a cold-start episode in which the steeper link seemed to protect against a bad extrapolation, dissolved under our own audit into a numerical artifact; explicit ridge regularization then delivered the genuine protection the artifact had only imitated (Section 6.3.1). Third, one anomaly survives every attribution attempt we make (Section 7).

Our contributions, in the order we would defend them: a methodology, the effect ceiling, that any researcher comparing two model families can compute tomorrow on their own data; a framework, the error budget, that turns "which model wins" into the more useful question "where does the error come from"; and a case study rigorous enough to trust, in which a large modern preference system is shown to be estimation-limited rather than approximation-limited, with every verdict pre-registered, thresholded, gated, and replicated.

2 Related work

The models we compare descend from a century of work on paired comparison. Thurstone modeled choices as comparisons of noisy latent performances [16, 21]; Bradley and Terry gave the logistic special case its modern form [2], which Luce showed is exactly the model implied by independence of irrelevant alternatives [15]. Harville and Plackett extended the family to multi-entry contests and rankings [11, 17], Davidson added an explicit tie outcome [6], and Elo and Glicko turned the ideas into the online rating systems used everywhere from chess to matchmaking [8, 9]. TrueSkill shares the Thurstonian generative story and fits it by approximate Bayesian inference [12]; we cite it for positioning and do not test it, for two concrete reasons. Its static link is the probit, the small-width limit of the lattice family we sweep, so its shape already lies inside our tested range; what distinguishes it is online approximate inference, and inference quality is a property of the fitting algorithm rather than the model family, which our design holds fixed precisely so that link shape is the only moving part. Ranking models without a latent performance scale, such as the Mallows family, sit outside the random-utility tradition entirely: they induce no link function for a ceiling to compare, and their behavior shifts delicately as the number of alternatives grows [1, 7]. The framework as stated does not cover them, a scope fact rather than a judgment.

The methodological ancestry of Section 4 also needs saying plainly, because none of its mathematics is new. The identity at the heart of the ceiling is the classical excess-risk decomposition of the logarithmic score from the theory of proper scoring rules [10]; the uniform-closeness bound is a cousin of the elementary divergence inequalities that relate closeness of distributions to indistinguishability of hypotheses [5, 13]; and verdicts framed as practical equivalence descend from the two one-sided tests tradition in bioequivalence [19]. What we believe is new is the placement: these quantities computed before either candidate is fit, on the empirical design distribution, as the opening step of a model-family comparison, together with the demonstration that a deployed system of current interest sits deep inside the regime they describe.

Cotton's lattice construction deserves a careful sentence, because we use its machinery while testing none of its main claims. The construction computes exact win probabilities for contests among any number of entrants, with arbitrary discretized noise and ties as dead-heat mass, and it remains coherent when entrants are added or removed [4]. That multi-entrant coherence is the paper's primary theorem. This paper is not a test of Cotton's primary theorem. Arena votes are pairwise, and on two entrants the lattice reduces to a one-dimensional curve in the ability difference, directly comparable to the logistic. What we evaluate is that curve and its tie mechanism. Testing field coherence itself would require listwise data that no public release contains.

On the platform side, the Arena leaderboard fits Bradley–Terry over deduplicated anonymous battles with ties given half weight in each direction [3]; our pipeline reproduces the published board to 0.18 Elo points of mean absolute error (Section 5). The platform's methodology has since grown style-controlled variants; we analyze the raw preference log and make no claims about style adjustment. A separate line of work documents violations of the axioms these models assume, from positional bias in LLM judging [20] to measured failures of transitivity and independence [22], with social-choice aggregation and listwise objectives proposed in response [14, 18]. The anomaly of Section 7 adds a human-vote observation to that literature: a violation shared by every static model we test rather than distinguishing between them.

3 Two ways to model a vote

Suppose model i has an ability θ_i, and a battle between A and B is decided by which side draws the higher performance $X = \theta + \varepsilon$, with independent draws of one shared noise distribution on the two sides. Only the difference between abilities matters, so any such model reduces to an increasing curve in the gap $g = \theta_A - \theta_B$:

$$\Pr(A \succ B) = F(g). \tag{1}$$

We call F the link. Bradley–Terry is the logistic link $\sigma(g) = (1 + e^{-g})^{-1}$, and it is the unique link consistent with independence of irrelevant alternatives. Continuous noise leaves no room for ties, and the two families we compare restore them in instructively opposite ways.

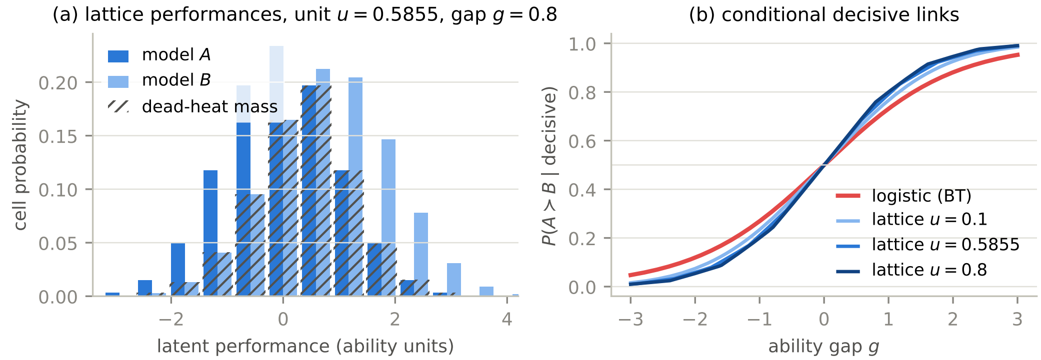

Figure 1: The lattice model of a vote. Panel (a) shows two discretized performance distributions at gap $g = 0.8$ and cell width $u = 0.5855$; a coarse lattice is drawn so the cells are visible, while fits use 500 cells per side. The hatched region is dead-heat mass, the probability that both performances land in the same cell. Panel (b) shows the decisive links this induces at three widths, next to the logistic. The lattice links are steeper at a toss-up, with slopes from 0.290 to 0.337 against the logistic's 0.25, and they steepen as the tie band widens.

The lattice family discretizes the noise onto cells of width u, so that performances take values $x_j = ju$. Writing $a_j(g)$ and b_j for the two sides' cell probabilities, the outcome probabilities are

$$W(g) = \sum_j a_j(g)\, \Pr(X_B < x_j), \qquad D(g) = \sum_j a_j(g)\, b_j, \qquad L(g) = W(-g), \tag{2}$$

and a tie is a dead heat: both draws in one cell, as in Figure 1a. The identity $L(g) = W(-g)$ rests on the shared noise distribution, which makes the difference of the two performances symmetric even when the noise itself is skewed. The width u is the family's single tie parameter, and we fit it by profile likelihood. Conditional on a decisive outcome the family predicts with

$$F_u(g) = \frac{W(g)}{W(g) + L(g)}, \tag{3}$$

and two properties of this curve shape everything downstream. It is steeper than the logistic at a toss-up, and it steepens further as u grows, because carving dead-heat mass out of the middle of the distribution sharpens what remains; the logistic's slope is not attainable at any width. It follows that a single parameter controls tie mass and decisive steepness together. The family cannot adjust one without moving the other, a property we will call entanglement and whose price we measure in Section 6.4.

Davidson's family extends Bradley–Terry instead. With tie parameter $\nu \geq 0$ and normalizer $Z_\nu(g) = 2\cosh(g/2) + \nu$,

$$\Pr(A \succ B \mid g) = \frac{e^{g/2}}{Z_\nu(g)}, \qquad \Pr(\text{tie} \mid g) = \frac{\nu}{Z_\nu(g)}, \qquad \Pr(B \succ A \mid g) = \frac{e^{-g/2}}{Z_\nu(g)}. \tag{4}$$

Theorem 1 (Tie orthogonality) *For every $\nu \geq 0$, Davidson's conditional decisive link is exactly logistic:* $\Pr(A \succ B \mid \text{decisive}, g) = \sigma(g)$.

Proof. Conditioning on a decisive outcome cancels the normalizer, leaving $e^{g/2}/(e^{g/2} + e^{-g/2}) = \sigma(g)$. □

The theorem makes the two families a matched pair in the cleanest possible sense. Each has one ability per model and one tie parameter. They differ in exactly one structural respect: Davidson's tie parameter never touches its decisive predictions, and the lattice's always does. Every tie-related result in this paper traces back to this contrast.

A few conventions used throughout. All fits are per-vote maximum likelihood with analytic gradients, and only the link differs between methods. Where an experiment mirrors the production leaderboard, decisive votes count twice in their direction and each tie counts once in each direction through the decisive link; where tie prediction is itself the question, we use the full three-outcome likelihood. Every fit anchors the same reference model, a dated checkpoint present with high volume in every training window, and ability magnitudes compared across links are rescaled so that one reported unit buys the same win-probability change at a toss-up in every method, the same role the familiar 400 over log 10 constant plays for Elo.

4 Effect ceilings and the error budget

The question "is the richer model better" has a prerequisite that is almost never computed: how much better could it possibly be? Both models will be fit to the same votes and scored on the same outcomes. If their links barely differ on the gaps the data contain, the comparison is over before it starts. This section turns that observation into a theorem, a number, and a framework.

Start with the honest version of the best case. Suppose the world really is the richer family: outcomes are generated by some lattice link F_u. The simpler family still gets to fit itself, so the richer family's true advantage is only what the best logistic concedes. With $\hat{P}$ the empirical distribution of fitted gaps, which does require one baseline ability fit before either candidate link is compared, that concession is

$$C \;=\; \min_{s>0}\; \mathbb{E}_{g\sim\hat{P}}\, \mathrm{KL}\big(\mathrm{Bern}(F_u(g))\,\big\|\,\mathrm{Bern}(\sigma(g/s))\big)\,, \tag{5}$$

and we call C the effect ceiling. The following theorem says the ceiling means what it appears to mean, and that a cruder version of it holds with no assumption about the world at all.

Theorem 2 (Approximation-limited regime) *Let p and p' be two prediction rules mapping gaps to win probabilities, and let $m(g)$ denote the smallest of $p(g)$, $1-p(g)$, $p'(g)$, $1-p'(g)$. (i) For every sequence of outcomes whatsoever, the difference in average log loss between predicting with p and with p' is at most $\mathbb{E}_{g\sim\hat{P}}[\,|p(g)-p'(g)|/m(g)\,]$. (ii) If outcomes are generated with true win probabilities $p(g)$, the expected excess log loss of predicting with p' instead of p equals $\mathbb{E}_{g\sim\hat{P}}\,\mathrm{KL}(\mathrm{Bern}(p)\,\|\,\mathrm{Bern}(p'))$. Consequently the expected advantage of the true family over the best member of a competing family equals the effect ceiling of Eq. (5), and predicting with any estimated member of the true family rather than the truth itself only reduces that advantage.*

Proof. For (i), fix one vote with gap g and outcome y. The prediction enters the loss as $\log p$ or $\log(1-p)$, and by the mean value theorem the two rules' losses differ by at most $|p-p'|$ divided by the smaller of the two arguments, which $m(g)$ bounds. Averaging over votes gives the claim. For (ii), take expectations of the log loss under $\mathrm{Bern}(p(g))$: the difference between using p' and using p is exactly the Kullback–Leibler divergence at each gap, and minimizing its average over the competing family gives the ceiling. Estimation error on the true side adds $\mathbb{E}\,\mathrm{KL}(\mathrm{Bern}(p)\|\mathrm{Bern}(\hat{p}))\geq 0$ to the true side's loss. □

Numbers need a unit, so we borrow one from Section 5 ahead of its derivation: the minimum practical difference for decisive calibration is 4×10^{-4} nats per vote, the cost of misreading every rating by ten Elo points, and we call one such quantity a threshold unit.

A toy example shows the computation at its simplest. Take the oldest rivalry in this literature, probit against logistic, and let gaps be standard normal, which is far more spread out than real matchups. The best-fitting logistic sits at scale 0.589 and concedes a ceiling of 0.67 threshold units. Even in a textbook world chosen to separate them, the two most famous links in statistics cannot be told apart at a level anyone would act on.

Now put the real numbers in. Arena matchups concentrate near toss-ups; the median fitted gap magnitude is 0.31, and every increasing link agrees at a toss-up by construction. On the observed gap distribution, the lattice link at its primary width and its best logistic competitor never differ by more than 0.011 in win probability, and on the average vote they differ by 0.0009. Part (i) of the

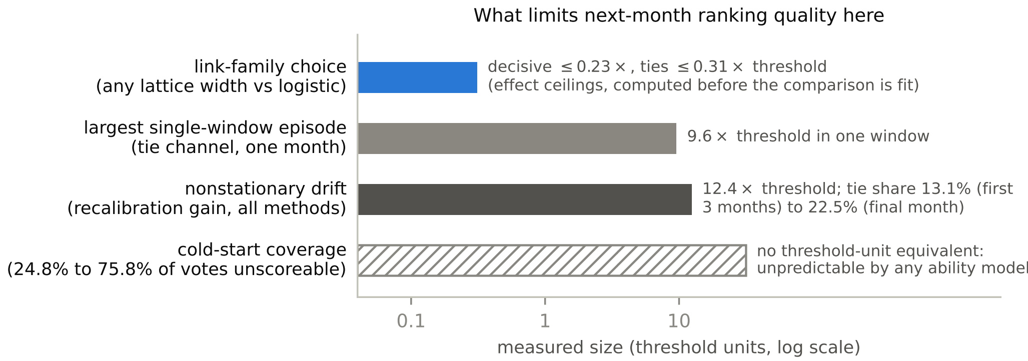

Figure 2: The error budget of the system we study. Bar lengths map to measured sizes in threshold units on a log scale for the three entries that admit the unit; the cold-start row has no comparable unit and is drawn hatched. Model choice is bounded by construction at a fraction of one unit, and every other line that admits the unit exceeds it by one to two orders of magnitude.

theorem, which assumes nothing about how outcomes arise, already confines any conceivable log-loss separation to 6.6 times the minimum practical difference defined in Section 5. Part (ii), the sharp in-family computation of Eq. (5), closes it to 0.009 at the primary width, and to at most 0.23 across every width and skew we consider plausible. The tie-mechanism analogue, computed on three-outcome predictions, is at most 0.31 in both truth directions. It is worth being precise about what that number bounds: the expected advantage of the truth over the competing family's best member, computed on the pooled gap distribution. A measured fit-versus-fit separation can exceed it when the fitted competitor is estimated rather than optimal, and when the world drifts so that neither family is the truth; and a single window whose gap distribution differs from the pooled one, a month of near-peer voting, say, is not bound by it at all. These are upper bounds on what fitting could ever recover under in-family truth on the pooled design, computed before either candidate was fit, and they set the honest expectation for everything in Section 6: practical equivalence on the decisive channel, and a tie channel whose one excess (Section 6.4) is of exactly the kinds just listed.

Three clarifications keep the theorem honest. Part (i) holds for every outcome sequence, dependent or adversarial included, because the bound is applied vote by vote; its only requirement is that both rules keep their predictions strictly inside the unit interval, which the factor $m(g)$ makes explicit. The large gap between 6.6 and 0.009 is the point rather than an inconsistency: the model-free bound charges every vote the worst of the two log derivatives, while the in-family computation weights each disagreement by how often the truth would expose it, so the two answer different questions, one ruling out large separations against any world and the other pricing the separation in the world most favorable to the richer family. And neither part is new mathematics; part (ii) is the standard excess-risk decomposition of the logarithmic score [10] and part (i) the kind of elementary bound information theory uses everywhere [5]. The contribution is where they sit in the workflow.

The ceiling is one line of a larger account. A ranking system's error has several possible sources: the model family may be unable to express the truth, which is approximation error; the data may be too thin or too new to pin down the parameters, which is estimation error; and the world may move between training and use, which is drift. Each source has its own instrument. The ceiling bounds the approximation line. Standard errors, coverage counts, and regularization experiments measure the estimation line. Recalibration slopes on held-out predictions measure drift. Filling in the lines for a given system is what we mean by an error budget, and Figure 2 shows the completed budget for Arena: the approximation line is bounded at a fraction of one practical unit, and every other line dwarfs it. The rest of the paper is the audit that fills in that figure, and we would argue the budget, not any single verdict, is the real product of the exercise. A reader comparing logistic regression to a neural network, or Plackett–Luce to Mallows, or Gaussian to Student-t noise, can compute the same two objects on their own data tomorrow, and will know before the comparison is fit whether it can matter.

5 The case study: data and protocol

Our main data is the largest public Arena battle log: 1,799,991 anonymous pairwise battles across 129 models, timestamped to the second, from April 2023 through August 2024. The platform's own deduplication retains 1,670,250 battles, and all fits use that subset. Outcomes are a win for either side, a tie, or a tie marked "both bad." Quality ties are 20.45 percent of the votes that are not both-bad over the full period, climbing from 13.1 percent in the first three months to 22.5 percent in the final month, a drift that will matter repeatedly. The replication set is the 2025 release: 135,634 battles across 53 models over fourteen weeks.

Before comparing anything we verified that the pipeline reproduces production. Refitting Bradley–Terry at the published leaderboard's own timestamp reproduces the published ratings to a mean absolute error of 0.18 Elo points with Spearman correlation 0.99997 across all 128 published models; the log's remaining model, days old at the snapshot, had not yet appeared on the published table. Extending the check to ten published snapshots spanning nine months, the era-appropriate variant achieves between 0.18 and 1.01 Elo points on nine of ten; the tenth (2024-04-03) reads 2.26, and diagnosis explains it. One model, five hours into the pool at the snapshot instant, carries a 72-point residual by itself, so the published snapshot and our archived log evidently disagree about that model's first hours of votes; excluding the single hours-old entrant brings the snapshot to 1.35, much closer to its neighbors' 0.18 to 1.01.

Verdicts need a threshold, because at a million votes statistical significance is free. We derive a minimum practical difference from a deployment-meaningful error. Leaderboard consumers read ratings at roughly ten-Elo-point granularity; a uniform ten-point error in every matchup translates to an expected excess log loss of 4×10^{-4} nats per vote at a toss-up, and that is our threshold for decisive calibration. The analogous computation for tie prediction, a one-point error in tie probability at the observed twenty percent tie rate, gives 3×10^{-4}. Differences below threshold are practically equivalent no matter how significant.

The ten-point anchor is a judgment, so we support it rather than assert it. The published leaderboard's own bootstrap intervals have a median half-width of 4.9 Elo points and a ninetieth percentile of 9.8, which puts a ten-point reading error at the edge of the board's own statistical resolution; differences below it are invisible to the platform's users twice over. The verdicts also do not depend on the exact number. Rerunning the calibration classifier at thresholds from five to twenty Elo points leaves every equivalence verdict standing, the interval stays inside the band down to a 4.3-point bar, and only an implausibly strict three-point standard flips anything (Appendix C). The tie comparison is the one place the threshold choice genuinely matters, and Section 6.4 says so where it does.

The protocol then fixes everything that could be fixed in advance. The full pipeline had to pass synthetic gates before touching real outcomes: it detected true effects down to a third of threshold with correct sign, called a positive only when one existed, recovered tie parameters within ten percent, and made no false calls in any world whose true effect was below threshold.

One gate result deserves its own paragraph, because it reverses a natural intuition and returns twice later. In a synthetic world with high parameter noise, the correctly specified steep link loses held-out log loss to a misspecified logistic by 1.41 threshold units: the steeper curve converts estimation noise into overconfident probabilities faster than its better shape earns credit back. Grounding the same computation in our data's calibrated standard errors, Fisher information deflated by the measured factor 0.80 and validated against the published bootstrap at correlation 0.983, shows the reversal is inactive at real pooled noise, where a true steep family would keep an advantage of at most 0.27 units. The mechanism returns as the natural reading of the small lean in Section 6.3, and as the intuition that the retracted episode of Section 6.3.1 appeared to confirm and did not. Real-data verdicts come from a fixed classifier over thirteen rolling monthly windows, training cumulatively and testing on the following month. A positive verdict requires the pooled effect to clear threshold, ten of thirteen windows to agree in sign, and a window-cluster bootstrap interval, whose effective sample

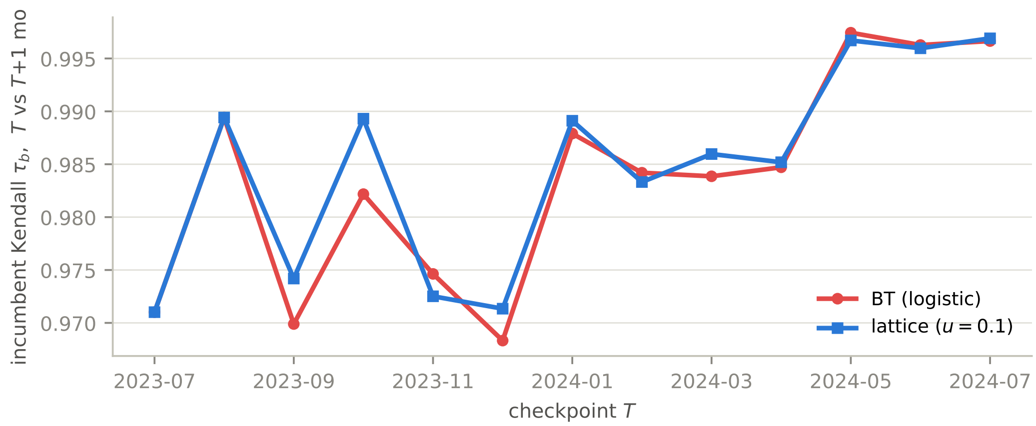

Figure 3: Month-over-month stability of incumbent rankings, measured by Kendall's τ_b, for the two methods across thirteen refit windows. The curves are statistically indistinguishable, and both improve together as the pool matures. Source: `rq1_metrics.csv`.

size is thirteen and is always stated, to exclude zero. Equivalence requires the interval to sit inside the threshold band, and significant leans below threshold are reported as leans, never as wins. Anything else is inconclusive and says so. Post-hoc analyses are labeled and never override a pre-registered verdict. Machinery ablations confirm the apparatus is not doing the work: pooled intervals move by under 0.004 threshold units across bootstrap sizes from one thousand to fifty thousand, link curves are invariant to lattice resolution at machine precision, and the profiled tie width moves by under one percent across profile grids. Finally, models absent from training are unscoreable and counted: they are 24.8 to 75.8 percent of next-month decisive votes, a number that belongs in the budget rather than a footnote.

6 What the audit found

6.1 Rankings do not move differently

The first worry about a growing pool is churn: new models enter, the leaderboard refits, and incumbents shuffle. If the logistic link were straining against reality, one might expect its refits to shuffle incumbents more than a flexible link's. So for each adjacent pair of monthly checkpoints we fit both methods on identical cumulative data and compare how each ranks the incumbents, defined as models with at least a thousand votes; an all-models variant behaves identically.

Nothing separates them. Quoting the $u = 0.1$ width, whose numbers the other widths repeat: the median paired difference in Kendall's τ_b is +0.00026, and the mean is +0.0011 with a window-bootstrap interval from −0.0001 to +0.0025 that touches zero; the lattice is more stable in seven windows, less in four, tied in two; the mean τ_b is 0.98358 for Bradley–Terry and 0.98469 for the lattice; the largest single-window difference is 0.00713. The picture repeats at all three lattice widths, at one-month and three-month horizons, and for both incumbent definitions. No incumbent moved more than five positions in any window under either method, and ability drift per month is 1.70 to 1.77 Elo-equivalent points for every method. The informative contrast in Figure 3 is vertical, not between the curves: stability itself climbs from τ_b of 0.968 to 0.997 as the pool matures, a swing twenty-six times the mean between-method difference and four times the largest single-window one. Stability is governed by sample size, not by link. That is the budget's first entry, and it is an estimation entry.

The ceiling predicted this: where the data live, the links barely differ. That prediction can be checked directly, without fitting any abilities at all.

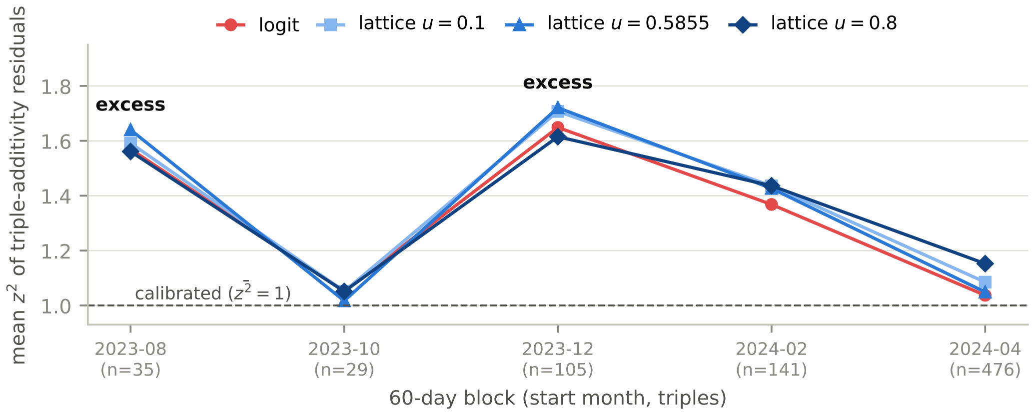

Figure 4: Mean squared additivity residual per sixty-day block, per link. The four links move together, inside the test's measured resolution, while two blocks, with a weaker echo in a third, show a shared excess that no link explains and that Section 7 takes up. Source: `rq2b_blocks.csv`.

6.2 The ceiling, confirmed without fitting

Any gap link makes a prediction that overdetermines the data. If the link is F, then for any three models the inverted head-to-head rates must add up: $F^{-1}(p_{AC})$ should equal $F^{-1}(p_{AB})+F^{-1}(p_{BC})$. How badly a triple violates this, scaled by its sampling noise, measures shape misfit in a way that treats every link fairly. The triples are not a clique of favorites, either: they span 43 models, with the five most-matched accounting for a third of triple memberships and the top ten for just over half. We compute the standardized residual

$$z = \frac{F^{-1}(\hat{p}_{AC}) - F^{-1}(\hat{p}_{AB}) - F^{-1}(\hat{p}_{BC})}{\sqrt{v_{AB} + v_{BC} + v_{AC}}}, \qquad v_{XY} = \frac{\hat{p}(1-\hat{p})}{n_{XY}\,[F'(F^{-1}(\hat{p}))]^2}, \tag{6}$$

for each of 786 dense stable triples, meaning all three pairings have at least a hundred decisive votes inside a sixty-day block and all three models are present throughout it. If the link's shape is right and preferences hold still, z is standard normal and the mean of z^2 is one.

The pooled means are 1.202 for the logistic and 1.231, 1.252, and 1.279 for the lattice widths. Synthetic calibration passed, with true links scoring 1.02 to 1.03, and the same synthetic runs measured the test's resolution: worlds generated by either family separate the four links by only 0.01 to 0.04, which is the same order as the real spread. So the test says what the ceiling said, through an entirely different instrument, one that involves no held-out scoring, no point estimates, and no time splits. At the gaps real matchups occupy, no member of this family is distinguishable from the logistic. Two independent derivations of the same bound are better than one.

Shape, then, is settled in-sample. Calibration on future votes could still differ, because estimation noise interacts with steepness even when shapes agree. That is the next entry.

6.3 Calibration: equivalent, and a cautionary tale

We fit both methods per window in the production tie treatment and score log loss on the following month's decisive votes, on the shared set both can score. The sign convention makes positive favor the lattice.

The verdict is equivalence with a lean. Pooled, the lattice loses by 0.121 threshold units at the primary width, by 0.148 and 0.226 at the others, with intervals inside the band and excluding zero, and with Bradley–Terry ahead in twelve or thirteen of thirteen windows. Every window sits inside the band; the largest is 0.63 (Figure 5). The lean is real, it is below deployment relevance, and per protocol we report it as a lean. It is largest at the steepest width, an ordering that holds in three

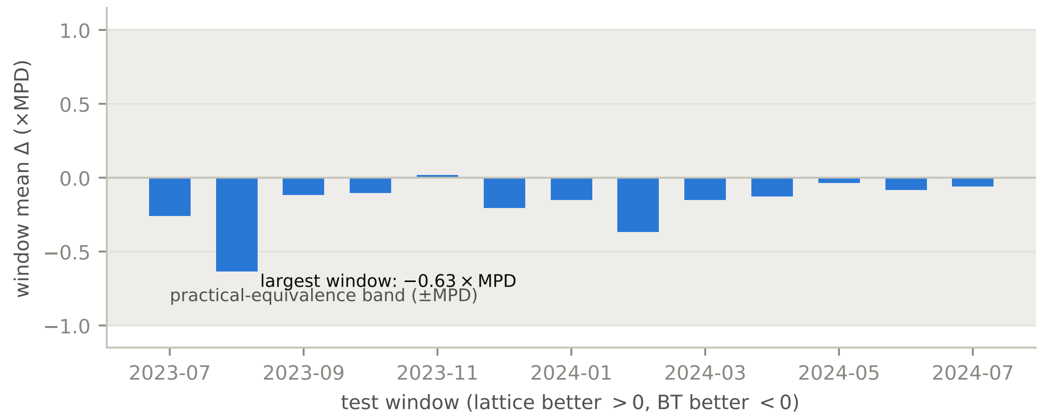

Figure 5: Per-window calibration difference at the primary width, in threshold units. All thirteen windows lie inside the practical-equivalence band. Source: `rq3_window_table_u0.5855.csv`.

quarters of window resamples, while the ordering of the two shallower widths flips in nearly half of them; we read the pattern as directional support for the overconfidence mechanism of Section 5 and the shallow-width inversion as noise. The pre-registered strata agree with the whole: recent entrants show the same sub-practical lean, and in the one stratum where the pre-analysis said a lattice-truth effect could clear threshold, the prediction of +1.58 is excluded decisively, with the measurement at −0.275 and an interval from −1.169 to +0.040.

Two shared measurements swamp all of the above, and they are the budget's drift and coverage lines. Every method's held-out predictions are underconfident by the same factor: recalibration slopes are 1.46 for all four models, and integrating parameter uncertainty into the predictions barely moves anything, shifting the pooled difference from 0.121 to 0.117 and the slopes from 1.455 to 1.459. The miscalibration is the world moving, not the estimator misfiring. And up to three quarters of next-month decisive votes involve a model with no training votes at all, which no link, prior, or smoothing scheme can score.

6.3.1 The episode that wasn't

Our first run of this experiment told a more exciting story, and the story was false in an instructive way. The original verdict was inconclusive: twelve quiet windows and one loud one, in which a model that had entered with two training votes was extrapolated to mid-table by Bradley–Terry while the steeper lattice held it near zero, and Bradley–Terry paid roughly 68 nats when the model was mass-sampled the next month and performed poorly. We read this as implicit shrinkage, a steep link quietly regularizing cold starts, and wrote it up as a mechanism.

The mechanism did not survive its own follow-up. Asked whether explicit ridge regularization could reproduce the protection, we ran a ridge sweep whose near-zero-penalty baseline refused to match our own stored table. The audit that followed located the cause in our earlier lattice implementation, which paired piecewise-linear log-likelihood values with smoothed gradients; the inconsistency, first noticed only as an optimizer nuisance, had been clamping near-zero-data models in the far tails of the curve. With consistent interpolation the episode vanishes. The old code reproduces the loud window exactly, at +5.792 threshold units; the corrected code puts it at +0.017; everything else is byte-identical. The corrected lattice places the cold-start model at +0.29, essentially where Bradley–Terry puts it. All synthetic gates were re-certified under the corrected code, and the equivalence verdict above is what remains.

The retraction sharpened the finding. Steepness is not implicit regularization. But explicit regularization does exactly what the false story promised: ridge on Bradley–Terry shrinks the cold-start model's estimate from +0.38 to zero as the penalty grows from one to three, improves that window

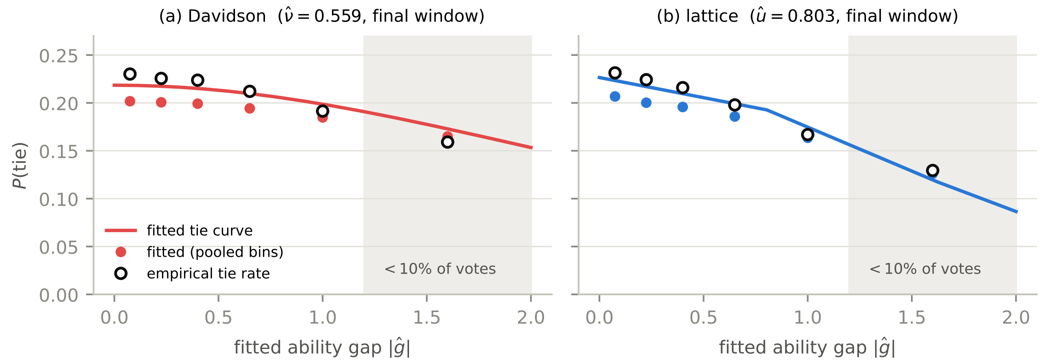

Figure 6: Fitted tie curves against empirical tie rates, pooled with vote weights, each model binned by its own fitted gaps. The mechanisms agree at the origin and decay at structurally different rates, and the region where they separate, shaded, carries under ten percent of votes. Source: `rq4_tie_curve_bins.csv`.

by about 5.5 threshold units, and moves established models by at most 0.015. Cold-start protection is real, cheap, and belongs to the estimation line of the budget, not to the model family. We kept the episode in the paper because it demonstrates the method: a pre-registered baseline and one skeptical follow-up caught an artifact that a single headline run would have published.

Calibration cannot separate the families either. By Theorem 1, ties are the one channel where they differ by construction, so ties get the last word.

6.4 Ties: different shapes, identical fit, and the price of entanglement

Both models now fit the full three-outcome likelihood per window, abilities plus one profiled tie parameter each, and are scored on the following month's wins, losses, and ties. Dead heats use the platform's quality-tie label; the both-bad label marks responses below an absolute bar rather than close draws, and appears only in the stress test below. Positive again favors the lattice.

The pooled verdict is inconclusive, and this time the episode is real data. The estimate is +0.838 threshold units with an interval from +0.017 to +2.542, the lattice ahead in ten of thirteen windows, and one window at +9.55 carrying the total. The estimate exceeds the pre-computed tie ceiling of 0.31, which is possible exactly because that ceiling binds in-family truth on the pooled gap design; the loud window is one month of near-peer voting that the pooled design does not describe, so the excess itself is evidence that the tie channel's error is drift, not link shape. That window decomposes cleanly: its tie votes contribute +407 nats to the lattice against −301 on decisive votes, led by near-peer pairs from the same families, the kind of sibling matchups where ties are common and the lattice's tie curve briefly paid off. A cold-start filter does not touch it, so this is one month of unusual voting rather than an estimation pathology; it immediately precedes one of the two anomaly eras described next, a coincidence we note and do not press. Robustness checks sharpen how much rides on single windows, in both directions: dropping the loud window turns the verdict into equivalence with a lattice lean of 0.28 threshold units, dropping one particular quiet window instead tips it positive for the lattice, and the other eleven leave-one-out verdicts stay inconclusive. This is also the only verdict in the paper that is sensitive to the practical threshold itself. At a stricter half-point tie-accuracy bar the lattice lean would clear significance, and at a relaxed two-point bar the comparison reads as plain equivalence (Appendix C). The tie comparison sits genuinely on the boundary, and one month of votes decides which side the point estimate lands on.

The structural result is sharper than the verdict. The two fitted mechanisms imply nearly identical tie probability at a toss-up in every window, within about 0.01 of each other, yet their bands have genuinely different shapes: the tie probability halves within 1.67 to 1.68 ability units under the lattice's overlap decay, against 2.84 to 2.93 under Davidson's slow decay, a stable factor of 1.7. Held-out fit

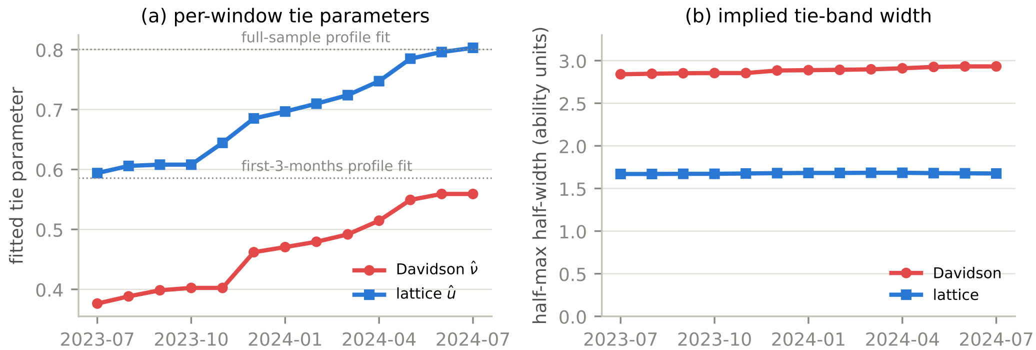

Figure 7: Tie parameters drift together. Panel (a): per-window profiled parameters rise as the platform's tie share rises, and the lattice width's endpoints reproduce two lattice profile fits computed independently months earlier, shown as dotted reference lines. Panel (b): the shape difference between the two bands is stable throughout. Source: `rq4_param_trajectories.csv`.

cannot tell them apart, with binned errors of 0.0225 and 0.0237, because votes concentrate where the shapes agree (Figure 6). The two tie parameters are not relabelings of one quantity. They carry different shape content that this dataset simply cannot adjudicate, which is a more precise statement than calling them equivalent.

Both tie parameters also drift, and the drift cross-validates. Over the sixteen months the profiled Davidson parameter rises from 0.376 to 0.559 and the lattice width from 0.594 to 0.803, tracking the raw tie share as it climbs from 13.1 percent in the first three months to 22.5 percent in the final month (Figure 7). Any single pooled tie parameter would average a moving target, which is why every experiment profiles per window. As an internal check, the windowed width trajectory's endpoints reproduce, to about 0.01, two lattice profile estimates computed months earlier by a different procedure on the early and full samples.

Finally, we tried to put a price on entanglement, and the honest report is that our first price did not survive its own stress test. Theorem 1 says Davidson's tie parameter cannot touch its decisive predictions and the lattice's must, so we forced both models to absorb both-bad-scale tie mass, about 35 percent of votes, and measured the damage on held-out decisive votes. The first run put the lattice's degradation at 4.75 threshold units against 2.50 for the Davidson control, an apparent entanglement cost of roughly 2.2 units. But the lattice widths in that run sat at the edge of their profile grid, and when we extended and refined the grid, every one of those widths moved to an interior optimum between 1.23 and 1.34: the apparent pinning was grid resolution, not a boundary. At the proper optima the lattice's decisive degradation averages 1.73 units, below the Davidson control, and no lattice-specific cost survives. What remains is the structural fact, which needs no measurement: the lattice's tie parameter provably steepens its decisive link while Davidson's provably cannot, so a deployment that cares about the decisive channel has one less thing to verify under the orthogonal design. We keep the paragraph because a theorem's empirical shadow failing to appear is itself information, and because this is the second time in the paper that stress-testing our own result changed the conclusion.

6.5 The same audit on a second dataset

Every design above transports verbatim to the 2025 release, with twelve weekly windows standing in for thirteen monthly ones; the transport itself was not pre-registered and is labeled accordingly. The full-population fits agree at Spearman 0.999839 with a maximum rank move of one. The calibration comparison again returns equivalence with a sub-practical lean toward Bradley–Terry at all three widths, at 0.084, 0.175, and 0.176 threshold units, with Bradley–Terry ahead in eleven of twelve windows. The tie comparison is again a lattice-leaning inconclusive, at +0.779 with an interval from +0.382 to +1.093. And the profiled width barely moves across fourteen weeks, from 0.689

to 0.693, which confirms the tie drift as a long-horizon phenomenon rather than a constant churn. The replication is more independent than the word usually implies: the 2025 pool shares exactly one model name with the earlier pool of 129, and its votes run thinner, with a median near 5,600 per model against 14,600. Different era, almost disjoint pool, same nulls, same lean directions, same structure.

7 The anomaly

One signal in this data survives every attribution we attempt, and it deserves its own section precisely because no model we test explains it. In two specific eras, with a weaker echo in a third, every link fails triple additivity the same way. The blocks beginning in late August and late December of 2023 show mean squared residuals of 1.57 and 1.65, with 13 to 14 percent of triples beyond two standard units against a nominal 4.6; the block beginning in mid-February 2024 is intermediate at 1.37, and the two cleanest blocks sit near one. Splitting blocks in half localizes the first episode to roughly late September and October and shows the second persisting undiminished in both halves, which rules out gradual preference drift within the window. We then checked the platform's deduplication transition, which overlaps only the cleanest block; entry activity, which is anti-correlated with the excess, since the loud blocks have fewer entrants and lower entrant vote share than the quiet ones; language mix, judge counts, votes per judge, tie share, and duplicated-prompt share, none of which separates loud from quiet; and concentration in particular models, which is absent. Because the excess is identical across link shapes, it is a property of the votes, not of any model.

In the units used elsewhere in this paper, the anomaly is not subtle at the level of individual triples: the typical additivity violation in the excess eras is 33 Elo-equivalent points against sampling noise of 45, compared with 25 against 36 in quiet eras, and the excess lives in the tails, where three times the expected share of triples land beyond twice their noise. The two surviving explanations would also leave different fingerprints, which we state so that better data can one day adjudicate between them. A shift in judge composition predicts an excess that tracks audience covariates, splits by judge cohort wherever identifiers exist, and strikes unrelated triples simultaneously. Genuine context dependence predicts an excess that attaches to particular pool configurations and enters and leaves with the models that create the context.

Two explanations remain live. The judge population may shift in composition, since a mixture of individually consistent judges violates additivity in aggregate. Or context may genuinely act on votes, with judgments depending on what else occupies the pool, which is the classical independence violation appearing as a shared failure of every model that assumes it away. Separating the two requires per-judge structure or listwise comparisons, neither of which public releases provide. We had pre-registered an event-study design for exactly this question and descoped it on evidence, since an instrument built on entry proximity is the wrong tool for a phenomenon anti-correlated with entries. We leave the anomaly as an open question, and as a concrete request to preference platforms: judge panel structure and occasional listwise comparisons would let someone answer it, and would also make the multi-entrant theory we deliberately did not test testable at last.

8 Reading the budget

The budget in Figure 2 is the paper's summary. The line that model choice controls was bounded by construction at a fraction of one practical unit, and sixteen months of audit plus a replication never found anything above it on the decisive channel, while the tie channel's one excess traced to drift rather than to link shape (Section 6.4). The lines that dominate belong to estimation and drift: a shared factor of 1.46 underconfidence on next-month votes, up to three quarters of votes unscoreable for want of any training data, single months in which one episode moves a verdict by ten threshold units, and a tie rate that drifts upward by nearly three quarters across the study period. A platform operator reading this budget knows the order of work, because each recommendation is a budget line read back. Cold starts first: shrink new entrants' estimates explicitly, since ridge buys the protection no link provides, worth about five and a half threshold units in the one natural experiment our data offered. Drift second: recalibrating predictions against recent outcomes is worth about twelve threshold units on our windows, roughly fifty times the entire link-choice ceiling, and parameter-uncertainty corrections do not substitute because the miscalibration is the world moving. Ties third, and only if the product cares: Theorem 1 favors the mechanism whose tie parameter provably leaves decisive predictions alone, even though we could not demonstrate the entangled design paying a measurable price here. And only after all of that, think about the link.

The generalization is the reason we wrote the paper this way, and it is a conjecture, so we frame it as one. Reward models for RLHF are Bradley–Terry by construction, and proposals to enrich the preference objective, listwise or otherwise, face the same arithmetic wherever comparisons concentrate near toss-ups and the candidate pool churns [14]. Recommendation and matchmaking systems consuming implicit pairwise feedback, sports and game rating systems, and econometric choice models fit to large observational panels sit in the same position: a simple link under theoretical suspicion, a richer family available, and a comparison that can be priced before it is run. We have demonstrated the budget's shape on one system; whether the same shape holds in those domains is exactly the question the method exists to settle, cheaply, before anyone commits to an expressiveness upgrade. What we would defend without hedging is the direction of the principle: expressiveness buys approximation accuracy, and buying it is only rational when approximation error is the binding constraint.

We would also defend the audit's conduct as a contribution. The ceilings set expectations before either candidate was fit. The synthetic gates certified that the machinery could detect what it claimed to detect. The fixed classifier refused two convenient readings. And the protocol caught our own false mechanism when one skeptical baseline declined to reproduce a stored number, which is exactly the failure mode that makes exciting stories survive into print. Negative results earn their keep when the method that produced them can be reused, and both halves of ours can.

9 Limitations

The disclaimers that matter most are scope. Nothing here tests the multi-entrant coherence property that is Cotton's primary contribution, because pairwise data cannot exercise it. Conclusions concern one platform, in two eras, and the gap-link families compared; TrueSkill-style approximate inference was positioned but not tested. The ridge demonstration rests on one natural experiment. The evidence for the overconfidence mechanism is directional rather than monotone. The empirical entanglement cost did not survive its own stress test and is withdrawn, which leaves the structural theorem without a measured counterpart here. The ceiling's gap distribution is itself estimated under a baseline ability model, a mild circularity that the width and skew sweep of Section 4 partially addresses. The method itself carries a requirement the case study made look easy: a defensible minimum practical difference. Arena offers one because its consumers read ratings at a known granularity and its published intervals set a floor; domains without a deployment-anchored threshold will find the ceiling easy to compute and the verdicts hard to define, and the budget framework inherits that difficulty. Finally, the anomaly of Section 7 is characterized but not explained. Resolving it, like testing field coherence, waits on data that platforms do not currently release: judge panel structure and listwise comparisons.

10 Conclusion

We set out to referee a model comparison and ended up pricing it. The price, computed before the comparison was fit, was a ceiling of a fraction of one practical unit, and the audit spent 1.8 million votes, a replication, four pre-registered comparisons, one retracted mechanism, and one open anomaly confirming that the price was right. The comparison itself is settled: for ranking language models from pairwise human votes, a richer Thurstonian link buys nothing that matters, and the system's real errors come from estimation, coverage, and drift. The method is not settled, and that is the point. Any proposed jump in model expressiveness can be priced the same way, with an effect ceiling on the family difference and an error budget for everything else, and the jump is worth making only when the ceiling clears the errors that remain. Spend expressiveness where the errors are.

Code and data availability

Everything behind this paper is public. The repository at https://github.com/Aviral1303/Thurstone-Models contains the full fitting and evaluation code, every pre-registration document with its thresholds and verdict criteria as they were fixed before fitting, the dated research log including the artifact investigation of Section 6.3.1, all result tables cited in the text, and the scripts that regenerate every figure from those tables. The battle data are the public Chatbot Arena releases; the conversion scripts in the repository download them directly from their original sources.

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A The interpolation artifact, in enough detail to avoid it

Section 6.3.1 retracts a mechanism because of a numerical bug, and a retraction is only worth as much as its diagnosis, so here is the bug precisely. Our lattice links are tabulated curves: the log-likelihood of an outcome at gap g is looked up from a grid of precomputed values, and its derivative feeds the optimizer. The original implementation interpolated the log-curve values piecewise linearly between grid points but supplied derivatives from a smoothed finite-difference table computed once on the grid. The two objects disagree: the derivative of a piecewise-linear function is a step function, not a smoothed one, so the optimizer received gradients that were not the gradients of the objective it was minimizing. Away from the tails the disagreement is negligible. In the far tails, where the lattice curve develops a staircase structure at probabilities near 10^{-4}, the mismatch let the line search terminate at points where the true objective still sloped, and models with almost no data, whose likelihood is nearly flat and tail-dominated, froze near wherever the optimizer first stalled. The fix is to make value and derivative come from one object: we now build a cubic Hermite spline of each log curve and differentiate the spline itself, so the pair is consistent by construction. A minimal reproduction needs only a two-vote model fit against a tabulated link with mismatched value and gradient tables; the estimate lands wherever the line search first stalls rather than at the maximum likelihood. Under the old code the affected window reproduces at +5.792 threshold units exactly; under the corrected code it is +0.017, with every other line of the pipeline byte-identical, and all synthetic detection gates re-certified after the change. The repository's research log records the discovery sequence, which began when a ridge baseline refused to match a stored table.

Quantity	Value	Source table
Battles (raw / deduplicated)	1,799,991 / 1,670,250	data audit
Pipeline replication of published board	MAE 0.18 Elo, ρ 0.99997	bt_replication
Validation across ten snapshots	0.18 to 1.01 Elo on nine	rq1_validation
Effect ceiling, decisive links	$\leq$ 0.23× threshold	rq3_ceiling_empirical
Effect ceiling, tie mechanisms	$\leq$ 0.31× threshold	rq4_tie_ceiling
Largest link disagreement on real gaps	0.011 win probability	theorem2_numerics
Stability, median paired $\Delta\tau_b$	+0.00026	rq1_metrics
Calibration verdict (primary width)	−0.121×, CI (−0.191, −0.074)	rq3_pooled_verdicts
Tie verdict (main)	+0.838×, CI (+0.017, +2.542)	rq4_pooled_verdict
Reliability slope, every method	1.46	rq3_reliability
Unscoreable next-month votes	24.8% to 75.8%	rq3_unscoreable
Recalibration gain	12.4× threshold	recalibration_gain
Ridge cold-start protection	~5.5× threshold	ridge_sweep_episode
Tie-band half-widths	2.84 to 2.93 vs 1.67 to 1.68	rq4_tie_band_table
Anomaly excess (loud eras)	mean z^2 1.57 and 1.65	rq2b_blocks
2025 replication, calibration	−0.084 to −0.176×	arena2025_verdicts

Table 1: Headline quantities with their source tables in the public repository. Threshold units are the minimum practical differences of Section 5.

B Key quantities in one place

C Sensitivity and robustness tables

Comparison	Threshold anchor	Pooled (×thr.)	Verdict
Calibration	3 Elo	−1.30	positive, Bradley–Terry
Calibration	5 Elo	−0.47	equivalence, lean BT
Calibration	7.5 Elo	−0.21	equivalence, lean BT
Calibration	10 Elo (paper)	−0.12	equivalence, lean BT
Calibration	15 Elo	−0.05	equivalence, lean BT
Calibration	20 Elo	−0.03	equivalence, lean BT
Ties	0.5 pp	+3.22	positive, lattice
Ties	0.75 pp	+1.43	positive, lattice
Ties	1.0 pp (paper)	+0.80	inconclusive
Ties	1.5 pp	+0.36	inconclusive
Ties	2.0 pp	+0.20	equivalence, lean lattice

Table 2: Verdicts under alternative practical thresholds (`mpd_sensitivity.csv`). The sensitivity pipeline computes each anchor exactly (4.14×10^{-4} nats at ten Elo points, 3.125×10^{-4} at one tie percentage point) where the headline verdicts use the rounded 4×10^{-4} and 3×10^{-4}; the paper-row estimates, −0.12 and +0.80, differ from the headline −0.121 and +0.838 by that threshold ratio alone. The calibration interval stays inside the equivalence band down to a 4.3 Elo bar. Leave-one-window-out checks (`loo_windows.csv`): all thirteen calibration verdicts remain equivalence; the tie verdict is inconclusive in eleven of thirteen, equivalence without its loud window, and positive for the lattice without one particular quiet window.